

Exploring equity standards for transport policy: A case study of calculated and perceived accessibility in the Greater Toronto and Hamilton Area, Canada

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Abstract

The lack of standards for the equitable distribution of accessibility complicates efforts to address transport poverty. This study identifies equity standards by exploring the relationship between calculated and perceived job accessibility for individuals living in the Greater Toronto and Hamilton Area, Canada. We use calculated accessibility indicators for three modes of travel (active transportation, public transit, and motorized modes) along individual-level variables to estimate ordinal models. To control for endogeneity we use spatial filters as instrumental variables. Two variables measure the perceptions of accessibility in the sample: the perception of having too few jobs available within an ideal commuting time and the perception of having to decline jobs because of transportation constraints. The results indicate that higher levels of active and transit accessibility tend to mitigate negative perceptions regarding these situations. Based on the results of these models, we identify accessibility thresholds that reduce the likelihood of negative perceptions for equity-deserving groups. We present three equity standards that we classify as conservative, intermediate, and progressive. We believe that these standards can serve as a data-driven example of how to support transportation planning in achieving more equitable outcomes, as well as for participatory processes to collectively define minimum or sufficient accessibility levels.

Keywords: benchmark, justice, justice, effect, egalitarian, sufficient

1. Introduction

Transportation equity refers to the distribution of the burdens and benefits of transportation systems among a population (Karner et al., 2020; Pereira et al., 2017). Burdens include health-related concerns arising from exposure to air pollution, noise, and traffic; sedentary lifestyles; and the risk of injury or death from collisions. In contrast, the primary benefit is the provision of accessibility (Pereira and Karner, 2021), understood as the ease of reaching (spatially distributed) opportunities and destinations (Páez et al., 2012; Hansen, 1959; Soukhov et al., 2025a), including employment, healthcare, educational services, and venues for social interaction.

Accessibility is one of the primary outcomes of spatial development (Páez et al., 2012). It is influenced by the spatial distribution of the population and opportunities and depends on the characteristics and performance of the transportation network. In other words, it is a product of both transportation systems and land use patterns (Wee and Geurs, 2011). There is a growing consensus among researchers and transportation planners that the goal of transport policy should be to improve accessibility (El-Geneidy and Levinson,

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2022; Boisjoly and El-Geneidy, 2017). This contrasts with traditional planning approaches that have prioritized mobility patterns that stimulate private automobile use and higher speeds as solutions to congestion and long commuting times. Mobility-centered policies often focus on metrics like increasing the number of trips, raising average speeds, and reducing congestion, treating mobility as an end in itself. However, this approach has proved problematic as movement becomes the measure of success, while leading to numerous other issues. In reality people travel to access opportunities, activities, or other people at their destinations, and so accessibility is seen as a superior indicator of performance (Pereira and Herszenhut, 2023).

Insufficient or inadequate accessibility can prevent individuals from participating in location-based activities they wish to engage in for reasons beyond their control, in a process described in the literature as Transport-Related Social Exclusion (TRSE) (Litman, 2024; Lucas, 2012; Preston and Rajé, 2007; Luz and Portugal, 2022; Paez et al., 2009). Insufficient accessibility, when combined with other forms of social and economic disadvantage, can result in transportation poverty (Lucas, 2019; Luz and Portugal, 2022): the failure of transportation and land use systems to provide access to essential services or employment. Transportation poverty increases vulnerability among individuals who already face challenges associated with age (van Dülmen et al., 2022), income (Allen and Farber, 2019), race (Awaworyi Churchill, 2020), gender (Murphy et al., 2023), disability (Upham et al., 2022), among other factors (Kim et al., 2024). With the suburbanization of low-income populations, it is estimated that over one million Canadians live in transport poverty (Allen and Farber, 2019).

Complicating efforts to address transportation poverty is the lack of standards for the equitable distribution of accessibility (Soukhov et al., 2025b). Additionally, there is an absence of transportation data in the form of open data products to support urban planners (Arribas-Bel et al., 2021; Behrendt and Sheller, 2021; Páez, 2021), and transportation planners receive limited formal guidance in assessing the equity implications of their projects (Miller, 2018; Palm et al., 2023; te Brömmelstroet et al., 2016). Furthermore, relatively few studies explicitly link accessibility levels with the experience of accessibility, such as employment opportunities declined or accepted due to transport conditions, or the ease to reach employment opportunities within ideal commuting times. Combined, this results in a gap in the development and implementation of accessibility standards to ensure equity in transportation planning and policy.

This study is guided by the following research questions: How does calculated job accessibility relate to perceived accessibility of employment opportunities and their outcomes? And could a given level of accessibility serve as a standard or goal for equitable transport planning? To answer these questions, this study has the general objective of proposing equity standards for transportation planning based on the relationship between calculated job accessibility and people’s perceptions of employment reachability.

Our research advances the literature on the equitable distribution of transportation benefits by demonstrating a model-based approach to identify calculated accessibility standards based on their relationship with perceived accessibility - more specifically, with perceptions about the ease to reach employment opportunities. This research is made possible by the release of Canada’s first National Survey of Transport Poverty (NSTP) (Tiznado-Aitken et al.). The proposed standards are derived at the neighbourhood level, while perceptions reflect the experiences of respondents to the NTPS that reside in those areas. In this way, these thresholds can assist policymakers in interpreting how different levels of calculated accessibility are associated with residents’ perceived ability to reach employment opportunities under existing social and spatial constraints. This increases the potential for government adoption and enhances the comprehensibility and relevance of such standards for community members and those experiencing unequal accessibility. Furthermore, identifying sufficient job accessibility levels can help assess and highlight areas at higher risk of transport poverty, particularly those with a higher concentration of equity-deserving populations.

In addition, to facilitate collaboration and further analysis and to ensure the transparency and replicability of our study, we developed this paper using literate programming, with the data analysis code accessible through our GitHub page (<https://github.com/dias-bruno/NSTP/>), in alignment with best practices in spatial data science (Páez, 2021; Arribas-Bel et al., 2021).

2. Theoretical Background

2.1. Calculated and perceived accessibility

The most commonly used accessibility measures are based on the gravity model of Hansen (1959) and make use of spatial data. This model considers the number of opportunities that a person can reach to be proportional to the sum of opportunities in each destination, discounted by the travel cost to reach that destination. This type of metric is classified as a location-based measure (Pereira and Herszenhut, 2023), representing a characteristic of a given location that assumes that all people in this location have equal access to the spatially distributed opportunities. However, they do not take into account the individual characteristics of people.

The perceived accessibility, on the other hand, refers to the perception of people of how easily they can reach opportunities (Lättman et al., 2016; Pot, 2025; Mehdizadeh et al., 2025), resulting from not only the interaction of land-use and transport infrastructure (Geurs and van Wee, 2004), but also from travel options (e.g., having a car or a transit pass), travel experiences, and individual characteristics and preferences. Often, perceived accessibility deviates from place-based calculated accessibility, with some research indicating that calculated accessibility might not be able to capture the multiple ways in which people experience a lack of access to opportunities (Zhu and Lucas, 2025). Explanations for these differences are partially attributed to differences in individuals' travel attitudes, capabilities, and preferences (De Vos et al., 2025; van der Vlugt et al., 2022).

Even though, as demonstrated in the following subsection, calculated accessibility is still commonly used to evaluate the effect that the interaction between land-use characteristics and the transportation system has on employment outcomes. In addition, existing studies on perceived accessibility seem to have difficulties in creating standardized definitions and measurement tools, contributing to the somewhat limited knowledge on the topic (Negm et al., 2025). In this context, connecting calculated to perceived accessibility helps to capture non-spatial and employment-related barriers that are not visible in place-based measures alone, since the calculated accessibility might be perceived differently even between people living within that same neighbourhood. In this paper, as suggested by Pot et al. (2025), we define sufficient levels of accessibility based on the relationship between calculated accessibility and people's perception of employment reachability, while also controlling for individual characteristics.

2.2. Job accessibility and employment outcomes

Studies connecting employment outcomes to job accessibility date back to the 1960s, when John Kain (1968) proposed the spatial mismatch hypothesis while analyzing Detroit and Chicago, USA. The author hypothesized that the low employment rates among Black Americans in inner cities resulted from employment decentralization and the difficulty faced by the Black population in moving out of these areas, primarily due to racial discrimination and segregation.

Since Kain (1968), studies linking calculated job accessibility and different employment outcomes have presented mixed results. Bania et al. (2008) found no significant effect for job access when examining its relationship with employment, earnings, hourly wage, and weekly hours worked for women leaving welfare in Ohio, USA. Delmelle et al. (2021) also found that changes in car accessibility had no significant impact on employment rates for different population income groups in the Metropolitan Area of Charlotte, North Carolina, USA, arguing that the spatial separation between lower-wage workers and their workplaces might already be relatively small. Studying Los Angeles, USA, Merlin and Hu (2017) found mixed-effects of job accessibility on employment status, with a stronger association for more highly educated groups. The authors argued that lower-educated groups may be more limited by non-spatial factors that impede their employment opportunities, such as qualifications and social skills. On the other hand, for workers with a higher level of education, the time and monetary costs of spatial separation itself may play a relatively more important role, so accessibility measures may have a greater impact on their decisions and ability to enter the labour market.

In contrast, other studies found significant effects of job accessibility on employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin and Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010). Jin and Paulsen (2018) used a difference-in-differences model for the Chicago Metropolitan Area, finding

that increasing job accessibility between 2010 and 2017 was associated with reduced unemployment rates. Duarte et al. (2023) found that job accessibility had a positive effect on participation in the labour market for women, a negative effect on informal employment for women, and a positive effect on the probability of being employed for both women and men. More recently, Bastiaanssen et al. (2025) examined differential employment effects of job accessibility by public transport in combination with the bicycle for different educational groups at the national level in the Netherlands. They found that jobs for lower-educated groups are often located outside areas of high accessibility, thereby reducing job accessibility among these groups, while jobs for higher-educated groups tend to be concentrated in and around downtowns and are associated with greater accessibility. In this study, the assumed connection between the spatial dimension of accessibility and the perceived employment outcomes is the basis for the identification of equity standards.

2.3. Equity Standards

An equity standard, when operationalized, defines the level of fairness to be adopted and the acceptable inequality (Soukhov et al., 2025b). Accessibility standards typically combine a concept of fairness with threshold criteria, reflecting both vertical and horizontal equity approaches. Vertical equity is often operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute walking trip for older adults), whereas horizontal equity commonly appears in the use of uniform travel impedance thresholds applied across all potential users (e.g., a maximum acceptable distance to a transit stop).

Yenisetty & Bahadure (2020) set an opportunity-based standard, considering a distance of more than 1,200 meters between amenities, services, and facilities by public transport to be unfair when analyzing cities in India. Tiznado-Aitken (2018) defined unfair cases as those where bus stops and metro stations are located more than 1 kilometer away. Sun & Thakuriah (2021) adopted 400 meters as the maximum acceptable walking distance to bus and tram stops, and 800 meters to ferry stations in England. In another study, Peunghumsai et al. (Peunghumsai et al., 2020) identified opportunity standards through the ratio of public transport supply to demand, considering conditions fair when demand is greater than supply.

The use of different thresholds reflects the idiosyncratic manner in which standards are developed, and the rationale for choosing a certain distance to, for example, bus stops or train stations, is often unclear. In addition, the use of distance-based thresholds may be problematic, since the time required to cross a certain distance can vary considerably within a city, even for the same mode of transportation. To address this problem, we suggest standards based on key thresholds of calculated accessibility.

3. Materials and Methods

Our approach for obtaining equity standards of accessibility based on the relationship between calculated measures and people’s perceptions can be synthesized in the following main steps: modelling job accessibility, predicting individual perception using calculated accessibility, setting equity standards for transportation, measuring transportation equity by using Foster-Greer-Thorbecke (FGT) measures (Foster et al., 2010), and defining priority geographical areas for policy interventions based on their Accessibility Fairness Index (AFI) (Martens, 2016).

3.1. Study Area and Data Sources

The study area for this research is the metropolitan region of the Greater Toronto Area and the Hamilton Area (GTHA), in Ontario, Canada. More specifically, the study area refers to the Census Divisions of the regional municipalities of Hamilton, Halton, Peel, Toronto, York, and the Durham subdivisions within the Toronto Census Metropolitan Area (Figure 1). According to the latest census (Canada, 2023), over 7.2 million residents live in this area, which is spread across around 7,300 square kilometres.

Regarding the primary data sources, this study uses data from two different sources: the latest Canadian Census (2021) from Statistics Canada; and the National Survey of Transport Poverty (NSTP) from the Mobilizing Justice Partnership (Tiznado-Aitken et al.). The Census was selected due to its various socio-economic and demographic variables and due to the “Commuting to Work” questionnaire, allowing the

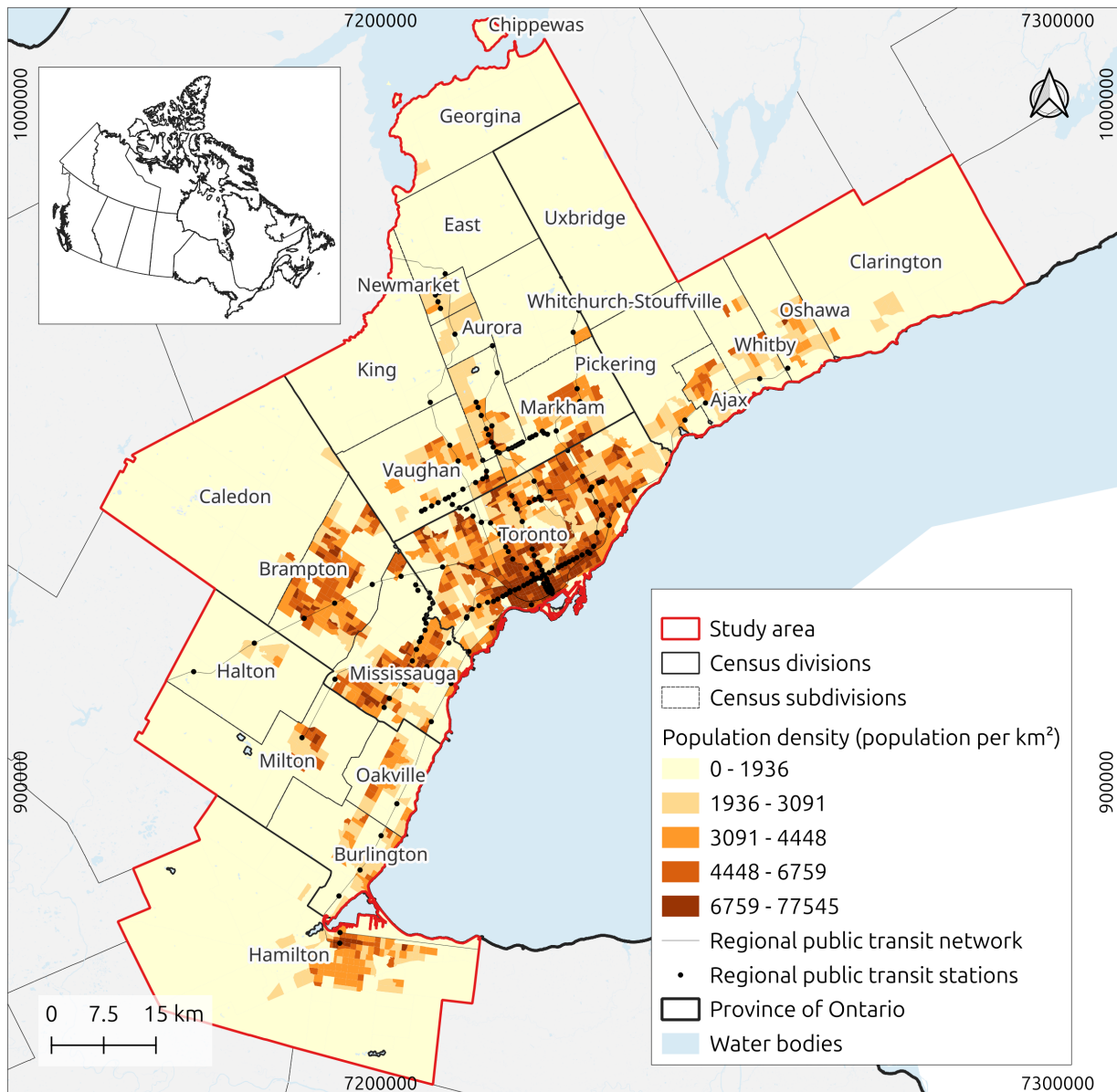


Figure 1: Map of the study area.

Table 1: Weighted summary statistics of the outcome variables.

To what extent do you agree with the following statements?	N = 3,576,893
There are too few jobs available for me in my ideal commuting time	58.84 (27.71)
I have had to decline employment opportunities because of my transport situation	41.00 (31.62)

¹ Mean (SD)

identification of individuals in the labour force and including variables required to calculate job accessibility, such as place of work, commuting modes, and commuting duration. The Census also provides standardized spatial datasets with statistical and administrative geographical boundaries, including the Census Tract (CTs), which are used in this study as the spatial units used for calculating accessibility measures (obtained using the **cancensus R package** (von Bergmann et al., 2021)). According to the Census(Canada, 2023), 3,868,120 individuals in the labour force reside in the study area, competing for a total of 3,043,730 job opportunities.

The second source of data, NSTP (Tiznado-Aitken et al.), is a large-scale survey designed to understand TRSE and transport poverty in Canada. It is a representative sample of more than 27,000 people across the country, and it provides an unprecedented perspective at the transportation challenges that Canadians face, especially among equity-deserving groups whose experiences are often overlooked. The survey is openly available to researchers, decision-makers, and the public, and offers information on household dynamics, respondents’ perceptions, and individual-level socioeconomic data. The NSTP provides respondent weights to enable the creation of statistics that are consistent and representative of the Canadian population. From this survey we select variables corresponding to respondents’ perceptions of accessibility and employment outcomes, in addition to socioeconomic and demographic variables at the individual level. The responses are geocoded by place of residence and can be assigned to Census geographies to link them to calculated accessibility.

Not all questions from the NSTP were answered by all survey respondents. The questions were defined according to demographic and socioeconomic profiles and to previous responses. For this study, the NSTP sample corresponds to respondents residing in the GTHA who also answered the questions displayed in Table 1, totalling 2,270 respondents that represent 3,576,893 individuals in the population (weighted sample). Using a continuum scale from 0 (completely disagree) to 100 (completely agree), the respondents answered to what extent they agreed with the statements in Table 1.

3.2. Calculating accessibility

In this study, we implement the multimodal spatial availability measure of Soukhov et al. (2024). This measure is a constrained indicator of accessibility that can account for multiple modes (see Soukhov et al., 2023, 2025a). This measure considers competition for opportunities, which prevents a single job opportunity to be available to more than one person. Competitive measures matter when considering employment outcomes because jobs are exclusive, and their use by one person prevents their use by another. In addition, this measure relies on a proportional allocation that balances the cost of travel and the population size, imposing a constrained calculation that allocates the opportunities to each origin. This ensures that, when all accessible jobs are summed, the sum is equal to the total number of opportunities in the region. Lastly, this measure is a more interpretable indicator since it returns scores in the units of opportunities.

We calculated the multimodal spatial availability V_i^m using Equation (1):

$$V_i^m = \sum_{j=1}^J O_j F_{ij}^{tm} \quad (1)$$

where:

- V_i^m is the spatial availability of opportunities at origin i and travel mode m ;

- O_j is the number of opportunities at destination j ;
- F_{ij}^{tm} is a balancing factor allocating opportunities at j to origin i considering mode m .

F_{ij}^{tm} incorporates both population demand and travel impedance, defined according to Equation 2:

$$F_{ij}^{tm} = \frac{F_i^{pm} \cdot F_{ij}^{cm}}{\sum_{m=1}^M \sum_{i=1}^N F_i^{pm} \cdot F_{ij}^{cm}} \quad (2)$$

The population-based allocation factor is obtained by:

$$F_i^{pm} = \frac{P_i^m}{\sum_m \sum_i P_i^m}$$

Where P_i^m represents the population at origin i using mode m . On its turn, the travel impedance allocation factor is given by:

$$F_{ij}^{cm} = \frac{f^m(c_{ij}^m)}{\sum_m \sum_i f^m(c_{ij}^m)}$$

Where:

- c_{ij}^m is the cost to travel between i and j for mode m ;
- $f^m c_{ij}^m$ is an impedance function representing the effect of travel cost.

Using microdata from the Census, we calculated for each CT the number of people in the labour force that commute to work by walking, cycling, motorized modes (those who commute by car, truck, or van as a driver or passenger, and motorcycles or scooters), and public transit (those who commute by bus, subway, light rail, streetcar, commuter train, and passenger ferry). We distributed remote workers according to the share of each transportation mode in each CT. Then, we summed the number of jobs in each CT considering the place of work mentioned by each worker.

We calculated the travel times between origins and destinations for each travel mode using the {r5r} R package (Pereira et al., 2021) with a street network obtained from OpenStreetMap. We used the General Transit Feed Specification (GTFS) obtained from the Canadian Public Transit Network Database (Canada, 2025) to retrieve the public transit schedule and coverage. Considering that our study area contains different transit agencies, we evaluated the GTFS files to identify a common date/week with public transit service. After this analysis, we set 8 a.m. (morning peak time) of December 22, 2025, as the trip departure time. Although this date falls during Christmas week, there were no differences in the public transit service schedules of the transit agencies due to the holiday, which makes this date representative of a regular weekday in terms of public transit travel time estimation. We also used Digital Elevation Models (DEM) from the Canadian Digital Surface Model (CDSM), with a spatial resolution of about 20 meters, to estimate the effect of declivity on walking and cycling times. We set a maximum duration of 180 minutes (3 hours) for motorized and public transit trips, with a maximum walking time of 5 minutes to reach the vehicle and 15 minutes to reach the public transit stop/station; for walking and cycling trips, we set a maximum duration of 90 minutes (one and a half hours). After estimating the travel times between CTs, we assumed that all intra-CT trip durations are equal to 0.1 minute.

Commute duration was used to calibrate Probability Density Functions (PDFs) to represent travel impedance, following the approach of Soukhov and Páez (2024), Verma and Ukkusuri (2025), and Dos Santos et al. (2026), where the impedance value represents the probability that the trip will occur given a specif travel time. We used the {fitdistrplu}s R package (Delignette-Muller and Dutang, 2015) and tested different functional forms (uniform, negative exponential, gamma, normal, and lognormal distributions) to select the best PDF for each mode of transportation and municipality, using a selection criteria the lowest Akaike Information Criterion (AIC) value (Akaike, 1974). The estimated impedance functions appear in Table A.1.

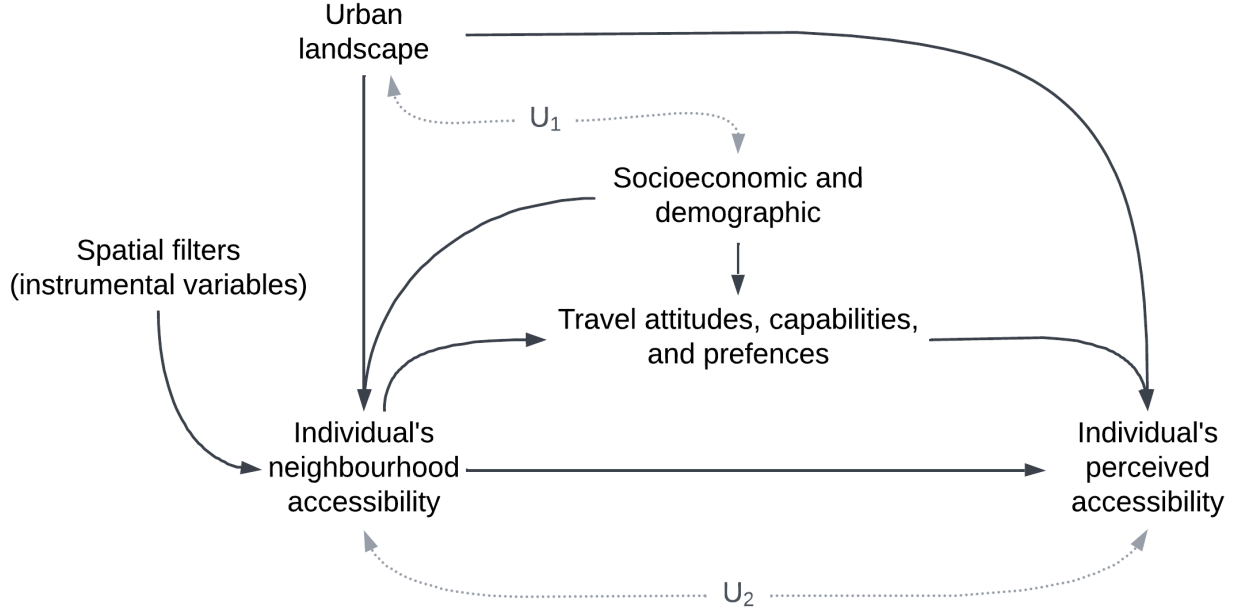


Figure 2: Causal diagram of calculated accessibility on perceived accessibility.

3.3. Modeling the causal relationship between calculated and perceived accessibility

In this study, we aim to define minimum levels of calculated accessibility by identifying key thresholds in its relationship with perceived accessibility. The underlying assumption of this strategy is that there is a connection between these two components of accessibility, such that changes in calculated accessibility would result on changes in the distribution of perceived accessibility. This implies a causal relationship between calculated (the cause or the treatment) and perceived accessibility (the outcome), rather than merely their statistical correlation. Therefore, we hypothesise that higher values of calculated accessibility will increase the probability of being in more positive perception categories, and that perceptions of accessibility can be connected to multimodal spatial availability indicators.

Following the steps suggested by Huntington-Klein (2021), we draw a causal diagram to graphically represent the assumed causal relationship between calculated and perceived accessibility (Figure 2). Causal diagrams show a chain of causal effects by using nodes to symbolize random variables and arrows to represent the causal effect between two random variables in the direction of the arrow (Cunningham, 2021).

According to our diagram, there is a direct effect of calculated accessibility on perceived accessibility. However, there are other ways in which they are connected. First, there are mediator components that influence the way a person perceives calculated accessibility. In our study, we use place-based measures to calculate accessibility, so that these indicators represent neighbourhood characteristics (CT level). However, each individual will perceive these neighbourhood characteristics differently based on their capabilities and preferences, as well as their travel options and experiences. For instance, a person with a disability that prevents or makes it difficult for them to walk or cycle will probably not enjoy the benefits of active accessibility to the same extent as a person without such a disability (Church and Marston, 2003); someone without a car will not take advantage of high car accessibility; someone without a monthly transit pass might take less public transit than someone with a transit pass, therefore reducing their potential for spatial interaction promoted by higher public transit accessibility.

In the causal diagram, the accessibility of a neighbourhood is caused by urban characteristics, such as land use and transportation infrastructure. In addition, place of residence, and consequently their neighbourhood accessibility, is affected (caused) by socioeconomic factors, with household income as the main driver of this

Table 2: Quintile thresholds used to categorize outcomes.

Outcome	20%	40%	60%	80%	100%
Few jobs available	35	52	69	87	100
Decline employment	8	26	51	72	100

decision, and also by demographic characteristics (such as gender, race, citizenship, among others). These socioeconomic and demographic factors not only determine where someone will live, but also shape how individuals experience travel and, consequently, make use of their neighbourhood accessibility.

However, it is also worth mentioning that socioeconomic and demographic factors and the urban landscape of place of residence are also linked through a common cause (U_1) that we cannot measure or properly define as a variable causing both. For instance, someone’s neighbourhood and their income level are probably related to generational wealth, a variable that cannot be obtained from the surveys. In addition, there must exist other factors, unknown or not translated into measurable variables, that probably affect both calculated and perceived accessibility (U_1).

From a causal diagram, we can identify all paths through which we can go from one node (a variable) to another using the arrows connecting them, with particular importance given to those paths connecting the treatment to the outcome. Paths in which all arrows point away from the treatment are front-door paths (Cunningham, 2021; Huntington-Klein, 2021); whereas paths that point towards the treatment are back-door paths. To identify the effect of the treatment on the outcome, we need to control for at least one variable in the back-door paths (hold this variable constant and unchanged), in a process that closes these biasing paths, while not controlling for at least one front-door path (Huntington-Klein, 2021). In this study, we employ regressions to estimate the relationship between treatment and outcome, controlling for all other measurable variables.

To model the causal diagram from Figure 2, we use weighted regressions that combine multimodal spatial availability per person with variables at the individual and household levels from the NSTP. Considering respondent weights, we transformed the perception variables into an ordered Likert scale based on their quintile position, categorizing them as: “Most Negative”, “Negative”, “Neutral”, “Positive”, and “Most Positive”. The thresholds used for this categorization is presented in Table 2.

Based on the causal diagram and on our literature review, we selected the following variables to represent each node:

- Calculated accessibility (treatment): multimodal spatial availability per person (active, public transit, and motorized modes).
- Perceived accessibility (outcome): individual perceptions categorized on a Likert scale, obtained from NTSP.
- Travel attitudes, capabilities, and preferences: most used transportation mode; whether the preferred transportation mode matches the most frequent mode; whether the individual has a monthly or annual public transit pass; number of cars and bikes in the household; and whether the person has any disability, all obtained from NTSP.
- Socioeconomic and demographic: whether the person is Indigenous or from a racialized population group; whether the person self-identifies as a woman; or as a transgender or non-binary person; age; household size and household income; and citizenship (temporary visa or refugee), all obtained from NTSP.
- Urban landscape: population centre class (rural, small, medium, or large urban population centre) and population density (dissemination area level), obtained from Census.

Considering that the outcome is an ordinal categorical variable, we used weighted ordinal logistic regressions (Lumley, 2011) to model the effect of the calculated accessibility indicators on the perception categories, using Equation 3:

Table 3: Weighted summary statistics of the explanatory variables.

Explanatory variables	N = 3,576,893
Active job accessibility per active traveller	0.06 (0.13)
Transit job accessibility per transit user	0.14 (0.09)
Motorized job accessibility per motorized user	1.00 (0.27)
Active transportation user	528,927 (15%)
Public transit user	829,940 (23%)
Person with disability	716,861 (20%)
Number of Cars	1.30 (0.92)
Number of Bikes	1.15 (1.33)
Preferred mode match most frequent mode	2,706,882 (76%)
Indigenous or racialized person	1,866,043 (52%)
Temporary visa or refugee	268,654 (7.5%)
Woman, transgender or non-binary person	1,757,263 (49%)
Age	40.15 (13.49)
Household size	2.90 (1.61)
Household income	
Less than \$40,000	561,182 (16%)
\$40,000 to \$80,000	1,145,633 (32%)
\$80,000 to \$120,000	930,128 (26%)
More than \$120,000	939,950 (26%)
Population density (people/km2)	7,809.82 (13,610.23)
Population centre class	
Rural	89,251 (2.5%)
Small	48,306 (1.4%)
Medium	58,128 (1.6%)
Large	3,381,208 (95%)
Public transit availability in the neighbourhood	3,435,002 (96%)
Montly or annual public transit pass	1,063,225 (30%)

¹ Mean (SD); n (%)

$$\log\left(\frac{P(Y \leq j)}{P(Y \geq j)}\right) = \theta_j - \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (3)$$

where Y represents the perception category of the respondent (ranging from “Most Negative” to “Most Positive”), and $X_1, X_2, \dots, X_p$ represent the set of explanatory variables, and θ_j and β_p are unknown parameters to be estimated. The parameters β_p are constant for all outcome categories and measure the change in the log-odds of reporting higher perception categories associated with each explanatory variable, while θ_j represents the intercept separating the perception categories. Table 3 presents the variables included in the regression models and their descriptive statistics (weighted mean and weighted standard deviation for continuous and dummy variables, and weighted count and percentage for categorical variables, created using the `{gtsummary}` R package (Sjoberg et al., 2021)).

3.3.1. Controlling for endogeneity

The presence of U_2 in our causal diagram implies the possibility of bias caused by omitted variables, in which unobserved factors simultaneously affect both calculated and perceived accessibility. Omitted variable

bias is the source of endogeneity, which occurs when an independent variable correlates with the regression error term and compromises the consistency of the estimators (Antonakis et al., 2010; Cunningham, 2021; Huntington-Klein, 2021). In this study, we address the potential endogeneity by applying an instrumental variable (IV) technique. An instrumental variable Z must be correlated with the outcome only through accessibility. Examples of instrumental variables used in studies analyzing the relationship between employment outcomes and job accessibility include Euclidean distance from the spatial unit centroid to the nearest major road or employment subcenter (Delmelle et al., 2021; Jin and Paulsen, 2018), share of automobile ownership (Hu, 2017), transit accessibility and car unavailability (Johnson et al., 2017), distance to rivers (Duarte et al., 2023; Luz and Portugal, 2022), and population density (Bastiaanssen et al., 2021, 2025).

However, two challenges arise in the search for relevant instruments: i) the likelihood that the instruments and error terms are correlated increases as the correlation between the endogenous variable and the instruments becomes stronger, which can render the instruments invalid; and ii) instruments that are only weakly correlated with the endogenous variable may be ineffective and perform poorly. In response to these challenges, Le Gallo and Páez (2013) developed an approach that relies on the use of synthetic instruments derived from the spatial configuration of the data (see Getis and Griffith, 2002; Griffith, 2003). Synthetic variables generated via spatial filtering ensure high correlation with the treatment and while remaining exogenous to the outcome.

Following the methodological approach of Le Gallo and Páez (2013) to create spatial filters for the accessibility variables:

1. We built a queen-contiguity matrix for the census tracts to represent spatial weights, expressing the existence of a neighbouring relation as a binary relationship (1 for neighbours and 0 otherwise).
2. Then, we computed all eigenvectors associated with the contiguity matrix. These eigenvectors are orthogonal to one another and represent a portion of the variance in the spatial contiguity matrix.
3. After this, we initialized a variable S with value 0 and linearly regressed each accessibility indicator (the independent variable) on each eigenvector and S . If the eigenvector coefficient was significant (adopting a p -value ≤ 0.05), we multiplied the eigenvector by its coefficient and added it to value of S . After this process, S represents the spatial filter associated with the accessibility indicator.

All spatial filters presented a Pearson correlation above 0.88 with their respective accessibility measures and absolute correlations lower than 0.15 with the original perception dependent variables (before transforming them into categorical variables). These results suggest that the spatial filters are strongly correlated with the endogenous accessibility variables while remaining weakly associated with the outcome, which is consistent with the relevance and exclusion conditions required for valid instrumental variables. In addition, we performed the Wald test to evaluate the null hypothesis of weak instruments and insufficient explanatory power to predict our endogenous variables.

Since our dependent variables are not linear, the popular technique of replacing endogenous variables by their first-stage predictors (two-stage least square estimation) is not appropriate - the so-called “forbidden regression” in econometrics (Wooldridge, 2010; Cunningham, 2021; Huntington-Klein, 2021). Therefore, we applied the two-stage residual inclusion (2SRI) approach (Terza et al., 2008), in which in the first-stage we estimated the accessibility measures using a linear regression including all control variables plus spatial filters (our instrumental variables). Next, in the second-stage, the dependent variables were estimated using the accessibility measures, the control variables, and the residuals from the first-stage regressions. In both stages, the respondent weights were considered to perform the regressions. In this approach, we use the regular treatment to predict the outcome, but also control for the unexplained part of the treatment.

3.4. Equity Standards

Considering only the cases in which the calculated accessibility was statistically significant, we measured changes in the predicted probabilities across perception categories by varying the levels of calculated accessibility, while controlling for the other explanatory variables. There are many possible ways of controlling the explanatory variables (e.g., fixing the number of cars in the household as 0, 1, or 2+; or fixing the respondent age as 30, or 31, and so on). Based on this, we decided to evaluate the effects of calculated accessibility in three distinct individual profiles:

- An equity-deserving profile, with individual-level control variables set to the values corresponding to populations facing the greatest transportation challenges.
- The most common profile, reflecting modal characteristics of the study area population and with control variables set to their most common values; and,
- An advantaged profile, defined by selecting for the most privileged population segment in terms of their transport experience.

For each individual profile, we increased the value of each accessibility measure by 0.01 jobs per person, while holding constant other accessibility measures (representing alternative transportation modes) at their weighted median values. We also adopted the weighted median value for the population density and a constant zero value for the residuals obtained from the first-stage regressions. For each transportation mode, we evaluated the effect of accessibility by considering the individual profiles as users of that specific mode. That is, for active travel accessibility, all individual profiles were set as active travellers, and so on for the remaining modes.

Then, for all individual profiles, we identified the minimum value of calculated job accessibility per person in which the probability of the positive and neutral perceptions combined was higher than the probability of being in negative perceptions, naming this minimum value as a conservative threshold. Next, we defined a progressive threshold as the minimum accessibility value that resulted in the probability of positive perceptions (not considering neutral perceptions) being higher than of negative perceptions. We then identified an intermediate threshold, set as the midpoint between the conservative and progressive thresholds.

Then, we selected as the equity standards the accessibility thresholds obtained based on the equity-deserving individual profile. The rationale behind this equity standard definition is to identify the minimum accessibility level that gives the most disadvantaged population a higher probability of having at least a neutral perception (conservative standard), a neutral to positive perception (intermediate standard), or a positive perception (progressive standard).

3.5. Measuring transportation equity

We calculated the extent and severity of accessibility gap using FGT measures (Foster et al., 2010), according to the Equation 4. FGT measures evaluates the number of people below a standard and the size of the accessibility shortfall relative to it.

$$FGT_{\alpha} = \frac{1}{N} \sum_H^{i=1} \left(\frac{z - y_i}{z} \right)^{\alpha} \quad (4)$$

Where N is the size of the total population; z is the accessibility standard; H is the number of individuals below the z ; y_i is the accessibility level of each individual i ; and α : determines how sensitive the index is to change in the accessibility gap.

Changing the values of α enables the calculation of different FGT indices (Karner et al., 2024). When $\alpha = 0$, FGT_0 measures the extent of undersupply as a simple headcount, indicating the proportion of people below the standard. When $\alpha = 1$, FGT_1 measures the severity of the undersupply as the average percent distance between the standard and the accessibility levels of the individuals below it.

3.6. Areas for priority intervention

For each equity standard, we identified CTs in the accessibility was lower than the equity threshold. Next, we calculated the AFI (Martens, 2016) to measure the severity of the insufficiency, by multiplying the population size below the standard by the accessibility gap. In practice, the AFI is a simple way to highlight areas with large groups of people that fall the furthest behind the set standards. After this, we classified CTs with accessibility shortfall into levels of priority for policy interventions: Very-high (AFI above the third quartile, top 25%), High (AFI between the median and third quartile, 25-50% from the top), Moderate (AFI between the first quartile and the median, 50-75% from top), and Low priority (AFI below the first quartile, bottom 25%).

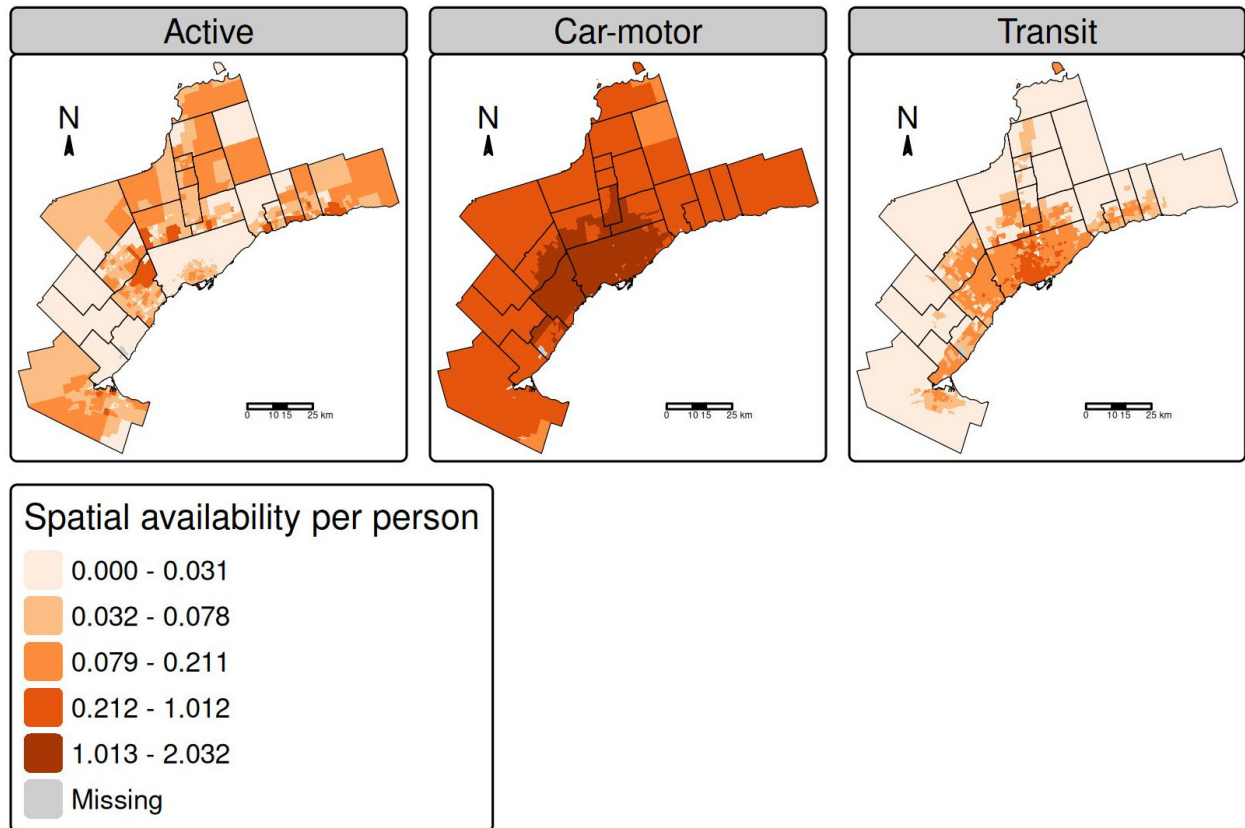


Figure 3: Maps of the multimodal spatial availability per person, considering active, public transit and motorized modes.

4. Results and discussions

4.1. Multimodal spatial availability per person.

Figure 3 presents the maps with the multimodal spatial availability of jobs per person. These maps show how, generally, motorized mode users can access more job opportunities, but also show that there is not much spatial variation when compared to the other two modes, with most of the CTs in the study area classified in the second-highest and or highest accessibility categories. Toronto and Mississauga present the highest level of motorized accessibility.

Transit accessibility appears to be more spatially distributed across the study area, with the city of Toronto presenting the highest rate of jobs accessible per transit commuters. Overall, we can observe an expected spatial pattern where CTs close to public transit infrastructure present higher levels of transit accessibility when compared to suburbs and rural areas, which often lack public transit services.

Next, the map of the active jobs accessible per active commuter (walkers or cyclists) does not present a clear spatial pattern, with many suburban and rural areas presenting higher levels when compared to more urbanized areas. This can be explained by either a higher number of jobs accessible within typical walking or cycling trip durations, less probable; or, probably, because the number of active commuters in these areas is very low, resulting in a higher rate of jobs per person, in a way that higher accessibility values in these areas may reflect low competition rather than higher absolute job availability.

Although motorized commuters represent about 78% of the labour force, they compete for 96% of the jobs in the study area. Transit users represent 15% of the labour force, but they reach only 3.4% of the jobs available, while the remaining 0.6% percent of jobs available are competed for by the 7% of active users.

On average, each active user respondent from the NSTP can access 0.06 jobs, while each transit user can access, on average, 0.14 jobs. These both measures are substantially lower than the number of jobs that each motorized commuter can reach, which is approximately one job per commuter on average.

4.2. *The effect of calculated accessibility on perceived accessibility*

Table 4 presents the two probability models linking calculated to perceived accessibility. We reject the null hypothesis of weak instruments and insufficient explanatory power to predict our endogenous variables (calculated accessibility indicators) based on Wald tests, given that all of the F-statistics are much higher than the Stock–Yogo critical values and are statistically significant ($p\text{-value} < 0.05$). However, the first-stage residuals were non-significant in the second-stage regressions, suggesting that there is little evidence of endogeneity in the calculated accessibility indicators, and that standard regressions (without the IV approach) could also have been appropriate. Despite that, we decided to continue our analysis using the models that account for potential endogeneity.

Calculated accessibility indicators were partially significant across the models. Motorized accessibility did not present significant values for any of the models, probably due to its ubiquity and homogeneity across the study area.

The model related to the perception of job availability (i.e., “there are too few jobs available for me in my ideal commuting time”) captures well the expected behaviour of accessibility, in which increases in calculated transit and active accessibility is associated with a higher probability of having a more positive of accessibility. In fact, this is the model in which the outcome is the most similar to the spatial availability of jobs definition, which refers directly to the number of jobs accessible, calculated using the impedance functions calibrated based on population behaviour. For this model, one additional job accessible by transit per transit user increases the average odds of being in a more positive perception group by 25%, indicating that fewer respondents agree that there are too few jobs accessible within their ideal commuting time. Active accessibility plays an overall significant effect for non-active travellers, reducing the odds of being in a more positive perception category by 14%. For active travellers, on the other hand, the interaction effect more than offsets this to result in a net positive effect. The same occurs in the model related to declining employment, whereby active accessibility is negative for non-active travellers, but positive for active travellers. We believe that the general negative effect of active accessibility for non-active commuters can be explained in terms of job competition, since increasing the number of jobs accessible for a mode that an individual does not use implies greater competition for jobs accessible through other modes, effectively reducing perceived accessibility for the mode a person actually uses. In addition, as previously shown in Figure 3, many of the areas with high active accessibility are located outside downtown cores, in more suburban areas with a low concentration of job opportunities. This further reinforces the assumption that some suburban areas exhibit high active accessibility not because of greater job availability, but due to a very low number of active commuters, resulting in a higher rate of jobs per person - reflecting low competition for jobs accessible by walking and cycling rather than higher active job availability.

Table 5 presents the control variable values applied for each individual profile, while Figure 4 displays the specific effect of increasing transit accessibility across the selected individual profiles for the “few jobs available” model. For all of them, we observe a decrease in negative perceptions as calculated transit accessibility increases. For the equity-deserving individual profile, the “Most negative” perception has high probability at low levels of calculated accessibility. In fact, at the median calculated transit accessibility (0.12 jobs accessible by public transit per transit user, calculated based on all NSTP respondents), the two most negative perception categories (“Most negative” and “Negative”) together account for over 52% of the total probability, resulting in a high likelihood that individuals in the equity-deserving group to have negative perceptions at this accessibility level. For the most common population, the initial pattern appears similar to the equity-deserving group, but the decline in negative perceptions occurs faster, with the “Most positive” probability surpassing negative probabilities at approximately 0.28 jobs per transit user. In contrast, for the advantaged population, we can see that even at very low levels of accessibility, the positive perceptions already present the highest probabilities and increasing transit accessibility only reinforces these positive perceptions.

Table 4: Weighted ordinal logistic regression models connecting calculated to perceived accessibility.

Variable	Few jobs			Decline employment		
	OR	SE	p-value	OR	SE	p-value
Active job accessibility per active traveller	0.86*	0.064	0.024	0.84*	0.085	0.033
Active mode interaction: accessibility $\times$ user	1.17*	0.072	0.029	1.21*	0.091	0.036
Transit job accessibility per transit user	1.25*	0.112	0.044	0.89	0.191	0.5
Transit mode interaction: accessibility $\times$ user	0.95	0.133	0.7	1.16	0.208	0.5
Motorized job accessibility per motorized user	0.93	0.097	0.4	1.17	0.147	0.3
Number of Cars	0.98	0.121	0.9	1.11	0.137	0.4
Number of Bikes	1.08	0.067	0.2	1.02	0.066	0.7
Preferred mode match most frequent mode	1.14	0.168	0.4	1.27	0.246	0.3
Montly or annual public transit pass	0.68*	0.174	0.026	0.52*	0.257	0.012
Public transit availability in the neighbourhood	0.75	0.245	0.3	1.04	0.228	0.9
Indigenous or racialized person	0.60**	0.164	0.002	0.57**	0.211	0.008
Temporary visa or refugee	1.61	0.257	0.064	0.70	0.421	0.4
Person with disability	0.69	0.209	0.077	0.80	0.232	0.3
Woman, transgender or non-binary person	1.05	0.149	0.7	1.02	0.195	>0.9
Age	1.01	0.006	0.080	1.02*	0.007	0.023
Household size	1.06	0.052	0.2	0.94	0.068	0.4
Household income			0.11	**		0.004
Less than \$40,000	—	—		—	—	
\$40,000 to \$80,000	0.80	0.188		1.50	0.272	
\$80,000 to \$120,000	1.23	0.220		1.99	0.250	
More than \$120,000	1.29	0.227		3.08	0.314	
Population density (people/km2)	1.18*	0.075	0.024	1.10	0.101	0.3
Population centre class			0.3	***		<0.001
Rural	—	—		—	—	
Small	1.34	0.487		0.86	0.484	
Medium	2.35	0.469		5.13	0.457	
Large	1.25	0.322		1.16	0.331	
Residuals: Active acessibility	1.09	0.078	0.3	0.97	0.085	0.7
Residuals: Transit acessibility	0.85	0.092	0.073	0.85	0.148	0.3
Residuals: Motorized acessibility	1.13	0.068	0.075	0.97	0.094	0.7

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001.

Table 5: Individual profiles to estimate probabilities.

Variable	Equity-deserving	Most common population	Advantaged group
Number of Cars	0	1	2
Number of Bikes	0	1	1
Preferred mode match most frequent mode	No	Yes	Yes
Indigenous or racialized person	Yes	Yes	No
Temporary visa or refugee	Yes	No	No
Person with disability	Yes	No	No
Woman, transgender or non-binary person	Yes	No	No
Age	35	40	45
Household size	3	3	3
Household income	Less than \$40,000	\$40,000 to \$80,000	More than \$120,000
Population centre class	Small	Large	Large
Public transit availability in the neighbourhood	Yes	Yes	Yes
Montly or annual public transit pass	Yes	No	No

Bold variables are statistically significant (p-value < 0.05) for at least one of the models.

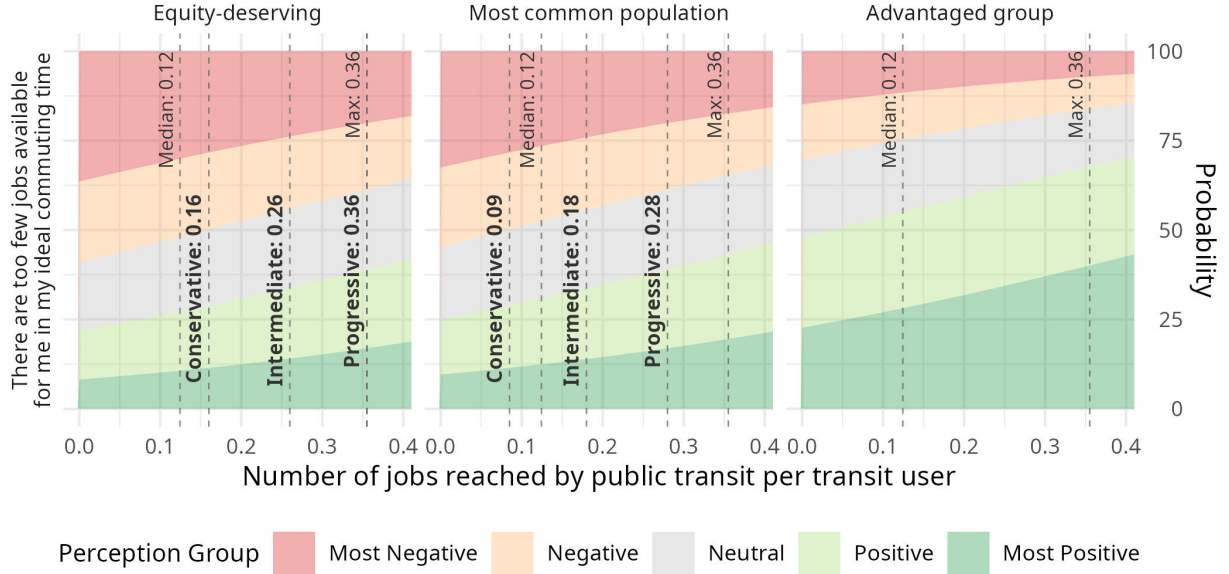


Figure 4: Public transit accessibility thresholds based on selected individual profiles.

Regarding the control variables, in the model related to “there are too few jobs available for me in my ideal commuting time”, having a monthly or annual public transit pass is associated with a 32% reduction in the odds of being in a more positive category. Being an Indigenous or racialized person is associated with a reduction of 40% in the odds of more positive perceptions, while increasing one person per square kilometres (population density) was associated with an 18% increase in the odds of being in a more positive category.

In the model related to “I have had to decline employment opportunities because of my transport situation”, having a monthly or annual public transit pass reduced the odds of being in a more positive category by 48%; being Indigenous or racialized reduced the odds by 43%; and each additional year of age is associated with an increase of 2% in the odds of being in a more positive group. In terms of income, higher income is associated with more positive perceptions (less perceived employment decline), with households earning more than \$120,000 CAD having approximately three times higher odds of being in a positive category compared to those earning less than \$40,000 CAD. In relation to the census population centre class, living in a medium-sized population centre is associated with substantially more positive perceptions (over five times higher odds), while living in a small centre is associated with 12% lower odds, and living in a large centre is associated with 16% higher odds - all this when compared to rural areas.

4.3. Equity standards and their impact on transportation equity

In this section we use the models presented above to identify active accessibility thresholds, while only the “there are too few jobs available for me in my ideal commuting time” model was selected to identify transit thresholds - since calculated transit accessibility was statistically significant only in this model.

Figure 4 shows the conservative, intermediate, and progressive public transit thresholds calculated for each individual profile. For the advantaged profile, a level of 0 jobs per transit user is already enough to result in positive perceptions, which means that the current transit accessibility levels would be sufficient and no individuals would fall below the minimum accessibility if this profile was selected as the reference.

For the most common profile, the conservative standard (the minimum level of accessibility at which the sum of negative perceptions is lower than the sum of all other perceptions) is equal to 0.09 jobs per transit user. This threshold is lower than the median calculated transit accessibility. The progressive

threshold, defined as the minimum level at which the sum of positive perceptions exceeds the sum of negative perceptions, for this profile is 0.28 jobs per transit user, with the intermediate threshold defined as the midpoint between these two values (0.18). However, for the equity-deserving profile, the conservative threshold is equal to 0.16 jobs per transit user, which is higher than the median accessibility level. The progressive standard is equal to 0.36 jobs per transit user, which corresponds to the maximum observed level of accessibility in the study area, and the intermediate threshold is 0.26.

Evaluating the thresholds for active accessibility, we again find that even a minimal level of accessible jobs per active commuter is already enough to result in positive perceptions for the most advantaged profile in both models, meaning that the current active accessibility levels would be sufficient and no individuals would fall below the minimum accessibility threshold. For the most common profile, the identified conservative threshold is also minimal; and the progressive thresholds are approximately 1.09 and 2.19 jobs per active user, resulting in higher probabilities of positive perceptions for the models of declining employment and few jobs available, respectively.

However, for the model related to declining employment opportunities due to transportation conditions, we were not able to identify a threshold that resulted in a higher likelihood of positive perceptions for the equity-deserving profile, even after estimating probabilities at more than double the maximum observed active accessibility level (2.03 jobs per active user). For the model regarding the perception that there are too few jobs available within an ideal commuting time, we identified 0.9 as the conservative threshold (22 times the observed median level) and a progressive threshold close to 5 jobs per active user (almost 2.5 times the maximum observed level and 123 times the median). These high thresholds are explained by the relatively small net positive effect of active accessibility, while the other explanatory variables exerted a greater influence on the estimated probabilities of belonging to a more positive perception group. Based on both models, the high active accessibility thresholds help illustrate the limitations of potential policies focused solely on increasing active accessibility to promote positive perceptions: an approach that intervene only in the spatial dimension of accessibility, without considering the capacities and travel preferences of the population, would likely result in limited impact on residents' perceptions regarding employment reachability and related outcomes.

These thresholds show how, depending on how the socioeconomic and transportation parameters are controlled and how the individual profiles are defined, different sufficient levels of accessibility are obtained. In fact, if one chooses to define thresholds based on the advantaged profile, it might lead to the conclusion that the current accessibility state is already sufficient and that no intervention or improvement is required. Alternatively, equity-deserving groups might still hold negative perceptions depending on the threshold based on the most common population is selected. Therefore, the only way to ensure that equity-deserving groups have a better chance of experience positive perceptions is to define minimum accessibility levels based on the controls tailored to these populations, setting socioeconomic and transportation parameters that represent them, thereby adopting a truly equity-based standard.

Considering the feasibility for policy interventions, we suggest the transit threshold defined in terms of the equity-deserving profile as examples of equity standards. Around 66% of the study area population lives in areas with lower transit accessibility levels than the conservative equity standard (0.16), while 91% live in areas with transit accessibility levels lower than 0.26 jobs accessible by transit per transit user (intermediate equity standard), and almost the entire study area population lives in areas below the progressive equity standard (0.36, which is also the maximum transit accessibility observed in the study area). In terms of the severity of transport disadvantage (FGT-1), the average shortfall among deprived individuals amounts to 36% under the conservative standard, meaning that those who do not meet the threshold fall short of it by roughly one-third on average (about 0.06). For the intermediate standard, the average gap is 53% (about 0.14). For the progressive equity standard, the average shortfall is about 0.23 (65% of the standard). These FGT measures show that, as the minimum transit accessibility levels become more demanding when moving from conservative to progressive standards, not only more individuals fall below the threshold, but those in accessibility shortfall are, on average, substantially further from achieving it.

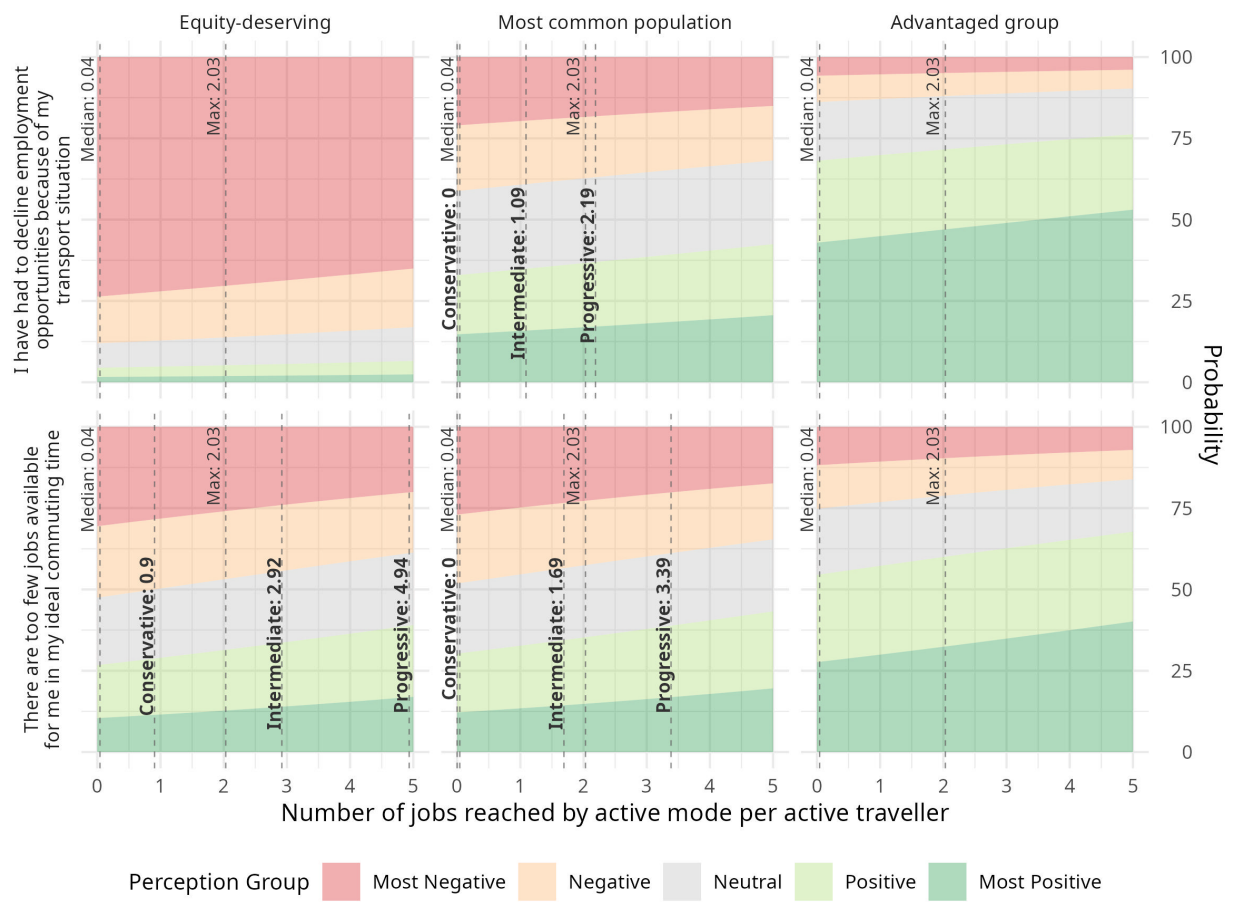


Figure 5: Active accessibility thresholds based on selected individual profiles.

4.4. Areas for priority intervention

Figure 6 shows the priority areas for policy intervention for CTs with transit accessibility shortfall, based on their AFI index and for each equity standard type. All census subdivisions contain CTs experiencing transit accessibility shortfall, independently of the type of standard. For the conservative standard, about 32% of the CTs (431 out of the 1,489 CTs in the study area) present transit accessibility levels above the shortfall limit, with most of them (91%) located in the City of Toronto. This share of CTs decreases to less than 10% for the intermediate equity standard (136 CTs). Only one CT located in Toronto does not present an accessibility deficit when evaluated under the progressive standard. The unique CT without accessibility shortfall corresponds to the area surrounding the St George Subway Station in Downtown Toronto, with a connection to the two subway lines that cross the city and connect neighbouring municipalities, running from North to South and from East to West. In addition, this CT is bounded by Spadina Avenue on the west and Bloor Street on the south, with many connections to the public transit system, including a streetcar line on Spadina.

In terms of areas for policy intervention, considering the conservative standard, the subdivision of Brampton is the one with the largest population living in areas classified as very high priority (217,253 people, representing 33% of the subdivision population), followed by Hamilton with 184,991 people (32% of its residents), Vaughan (144,670 people, 44%) and Markham (142,066 people, 41%). For the most progressive standard, Brampton and Hamilton again appear having the largest populations living in very high priority areas, with 380,289 people (57%) and 229,549 people (40%) respectively, but now followed by the City of Toronto with 221,768 people (7%).

However, in terms of population share, 100% of the population living in the subdivisions of East Gwillimbury (34,637 people), Uxbridge (21,556 people), and Whitchurch-Stouffville (49,864 people) are in areas with very high priority for policy intervention for the conservative and intermediate standards. For the progressive standard, only Whitchurch-Stouffville has all its CTs classified as very high priority areas. This difference occurs because the priority areas are defined based on their AFI position in the quantile classification for the entire study area, and as additional CTs are included under more rigorous standards, the AFI distribution also shifts. Regarding the Whitchurch-Stouffville, this subdivision is served by only one fixed public transit line managed by GO Transit, a regional train line that connects Stouffville with the subdivisions of Markham and Toronto. The list with the total number of people living in each priority area, by equity standard type and subdivision, is available as supplementary material.

5. Critical analysis and limitations

Although the unprecedented look at transport poverty in Canada provided by the NSTP, the survey does not contain a question in which the respondent answers objectively whether they are satisfied or not with their mobility experiences or overall transport conditions. Rather, the respondents use a continuum scale to answer their level of agreement with specific statements. The absence of a categorical or binary variable referring to satisfaction level led us to categorize the outcomes, in order to identify which perception category the respondent belongs to, varying from most negative to most positive. In fact, with the information we have, we cannot infer if the person is satisfied or not, but rather whether they are more or less satisfied in comparison to others. The underlying rationale for our standards proposition is to identify the calculated accessibility level that places the most disadvantaged population group within the most positive perception group.

It is worth mentioning that other outcome categorizations or numbers of categories are possible. For instance, one might decide to convert the continuum scale to a Likert scale based on fixed intervals, classifying those with agreement levels between 0 to 20% as “Most Negative”, 20 to 40% as “Negative”, and so on, in pre-determined intervals. The problem with this approach is that, when doing this, the analyst is normatively defining levels of satisfaction for the respondents that might not represent the actual feeling of the person. Because of this, we decided to categorize responses based on the perception scale distribution and use the sample population to evaluate those who are more and less satisfied.

This outcome categorization caused, in general, the control variables to lose part of their predictive power when compared to linear regression models created using the previous continuous forms. As shown in Table

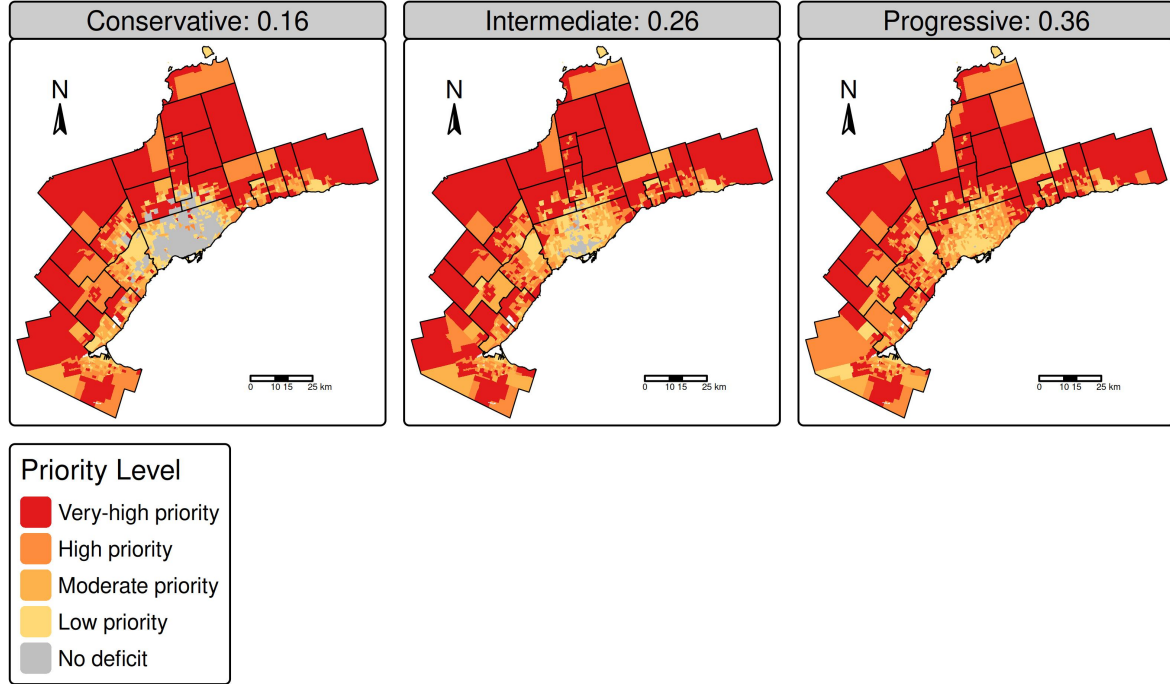


Figure 6: Priority areas for policy intervention by type of equity standard.

A.2, in general, the accessibility variables presented a similar behaviour when compared to the categorical models; However, the control variables presented a better performance in explaining the outcomes in the linear regressions, such as in the case of the disability indicator and the “temporary visa or refugee” variable, which did not present statistical significance in the categorical version of the model for “there are too few jobs available for me in my ideal commuting time.” We decided to use the categorical models because they enabled us to identify levels of satisfaction in a non-normative way.

Regarding the relationship between perceived and calculated accessibility, only transit accessibility, in one of the models, presented the behaviour hypothesized in which increasing accessibility was associated with more positive perceptions. Although the active accessibility indicator presented a slightly positive net effect for active commuters in both models, this effect was not sufficient to outweigh the influence of the other explanatory variables. As a result, the estimated active thresholds were excessively demanding and unrealistic from a policy perspective. For one model, we were unable to identify thresholds, while for the other, the conservative threshold was 22 times higher than the median accessibility level and the progressive threshold was more than 123 times higher than the median accessibility. These results led us to propose equity standards based solely on the model “there are too few jobs available for me in my ideal commuting time,” and for public transit mode.

For future research, we suggest exploring additional place-based accessibility measures to further evaluate the relationship between calculated and perceived accessibility. For instance, the raw multimodal spatial availability indicator could be tested. This indicator uses as output the number of jobs accessible by different transport modes, not standardized by the number of people using these modes. We decided to use multimodal spatial availability per person because we believe that competition matters when evaluating perceptions related to employment outcomes, as demonstrated by previous studies (Jin and Paulsen, 2018; Merlin and Hu, 2017; Miller, 2005). Another limiting factor that led us to adopt the CT level was that we could not estimate travel times at the DA level for the entire study area due to the memory limitations of our

computational resources. Despite that, the initial tests performed with a smaller study area did not show differences in terms of the sign and statistical significance of the accessibility coefficients between regressions - probably because the indicators are standardized.

Continuing with the spatial unit used, the CTs, we recognize that this unit is not the most detailed among census geographies, being less detailed than dissemination areas or dissemination blocks. We decided to use CTs due to the large size of the study area, in order to provide a regional analysis rather than focus on only one city, considering this entire basin to calculate the accessibility indicators and select the sample. This choice is motivated by the recognition of commuting flows across municipalities, which are even one of the parameters used by Statistics Canada when defining Census Metropolitan Areas.

Regarding the equity standards definitions, three moral principles guided our approach. The first relies on the recognition of accessibility as a human capability (Sen, 2005, 1999; Pereira et al., 2017; Luz and Portugal, 2022), something required for a dignified life (Karner et al., 2020), and the main transportation benefit to be distributed across society in order to increase people's freedom. The second is grounded in the sufficientarian perspective, considering a fair transportation and land-use system as one that provides sufficient accessibility to all members of society (Martens and Golub, 2021) as requirement for them to meet their basic needs (Pereira et al., 2017; Pereira and Karner, 2021). The third moral principle is grounded in an egalitarian perspective, in which the minimum level of accessibility is defined in terms of the most equity-deserving population groups to reduce inequalities of opportunities. Making explicit the moral foundations that support our analysis is, in our perspective, important because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits.

Transport equity is only one step towards achieving transport justice (Karner et al., 2020): a condition in which no person or group is disadvantaged by a lack of access to the opportunities they need to lead a dignified and healthy life. To achieve this state, it is necessary to transform the structures and processes that lead to the unequal distribution of the multiple externalities of transport across population groups and space. In this study, we propose a methodology to suggest equity standards based on a data-driven approach. We intend to suggest minimum levels of accessibility that could guide transportation planning towards a more equitable distribution of transportation benefits, as an earlier stage to be succeeded by a participatory and democratic process to collectively select the best standard, considering that residents and other stakeholders must be able to actively participate in and influence the decisions that affect their lives. While this is, in our perspective, the most appropriate method for standards definition, it is beyond the scope of this study to develop democratic plenaries to discuss the suggested standards.

A positive aspects from this research is that the equity standards are suggested based on their link with perceived accessibility, making them easier to communicate with policy practitioners and the general public. We also accounted for possible endogeneity of calculated accessibility, so that the coefficients represent the true effect of the indicators and not a mere a correlation. Although the study does not ensure full reproducibility, since the used surveys are confidential, it ensures replicability by making all codes available for consultation.

6. Conclusions

The main objective of this study was to suggest equity standards for transportation planning based on the relationship between calculated job accessibility and perceived accessibility referred to employment outcomes, for a study area covering municipalities around the cities of Toronto and Hamilton, southern Ontario, Canada. For that, we used data from the 2021 Census of Population and the recently released National Survey on Transport Poverty (Tiznado-Aitken et al.), combining place-based calculated accessibility measures with variables at the individual level, while controlling for endogeneity using instrumental variables in a two-stage regression estimation.

The place-based accessibility models were created using multimodal spatial availability per person, considering walking and cycling (later combined as active transportation), public transit, and motorized transportation modes. In general, our models showed that these indicators partially predicted with statistical significance perceptions related to how easy it is to reach work and declining employment opportunities

because of transport situation. For the model regarding the perception of having too few jobs available within an ideal commuting time, public transit accessibility behaved according to our hypothesis that increasing calculated accessibility would reduce negative perceptions, in which one additional job accessible by public transit per transit user increases the average odds of being in a more positive perception group by 25% (p-value < 0.05). Our models also showed (p-value < 0.05) that, for active users, active accessibility has a slightly positive effect on individual perception, increasing their odds of being in a positive category. Using the “few jobs available” model and for the public transit mode, we suggest equity standards based on the accessibility threshold that, results in a higher likelihood of positive perceptions for people from equity-deserving groups. We suggest three equity standards: a conservative standard (0.16 jobs reached by public transit per transit user), an intermediate standard (0.26), and a progressive standard (0.36).

Our results show that most of the study area falls short of the proposed equity standards, regardless of their category (conservative, intermediate, or progressive). About 66% of the study area population lives in areas with transit accessibility below the conservative standard, 91% in areas below the intermediate standard, and almost the entire study area lives below the progressive standard.

Finally, we believe our study contributes to the ongoing discussion of equity in transportation, and that equity standards based on, or inspired by, our methodology can serve as a data-driven starting point for debate among residents and other stakeholders within a participatory and democratic process to collectively define the most appropriate standard.

CRedit authorship contribution statement

(include after the review)

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Data availability

We created this paper using literate programming in which the R markdown code to fully reproduce this article is available on our GitHub repository (<https://github.com/dias-bruno/NSTP/>).

Declaration of competing interest

The authors declare no conflicts of interest.

Appendix

Table A.1: Impedance functions calibrated for the census subdivisions of the study area.

Subdivision	Census Code	Mode	Impedance function	Parameter 1*	Parameter 2*	AIC
Durham	3518	Car-motor	Gamma	1.92	0.07	1714717
	3518	Walk	Exponential	0.09		42507
	3518	Transit	Gamma	3.33	0.06	121000
	3518	Bike	Gamma	1.89	0.08	3829
York	3519	Car-motor	Gamma	2.16	0.08	2549528
	3519	Transit	Normal	54.89	28.44	200698
	3519	Walk	Exponential	0.09		58149
	3519	Bike	Lognormal	2.75	0.81	8060
Toronto	3520	Transit	Gamma	3.60	0.08	1880937
	3520	Car-motor	Gamma	2.33	0.09	3942809
	3520	Walk	Gamma	1.33	0.08	449370
	3520	Bike	Lognormal	2.94	0.65	123658
Peel	3521	Car-motor	Gamma	2.33	0.09	3351993
	3521	Walk	Exponential	0.08		72360
	3521	Bike	Lognormal	2.87	0.73	8853
	3521	Transit	Gamma	3.28	0.06	468686
Halton	3524	Car-motor	Gamma	1.92	0.07	1264978
	3524	Transit	Normal	53.20	26.65	75851
	3524	Walk	Lognormal	1.90	1.08	40360
	3524	Bike	Gamma	1.99	0.11	6229
Hamilton	3525	Car-motor	Gamma	1.85	0.07	1323879
	3525	Walk	Exponential	0.08		61816
	3525	Transit	Gamma	2.24	0.05	128266
	3525	Bike	Lognormal	2.83	0.75	9830

Note:

For each probability distribution, Parameter 1 and Parameter 2 correspond to their standard defining parameters: the mean and standard deviation (for lognormal and normal), the rate and shape (for gamma), and the power (for exponential). The AIC represents the Akaike Information Criterion used to assess model fit.

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Table A.2: Weighted linear regression models connecting calculated to perceived accessibility.

Variable	Few jobs			Decline employment		
	Beta	SE	p-value	Beta	SE	p-value
(Intercept)	79***	7.42	<0.001	50***	9.31	<0.001
Predicted active accessibility per user	2.1	3.10	0.5	13**	4.61	0.006
Predicted transit accessibility per user	-76***	18.2	<0.001	-5.1	26.3	0.8
Predicted motorized accessibility per user	-6.6	5.20	0.2	-9.2	7.05	0.2
Number of Cars	0.65	1.54	0.7	0.20	1.62	>0.9
Number of Bikes	-1.5	0.947	0.11	-0.16	1.06	0.9
Preferred mode match most frequent mode	-0.29	2.10	0.9	2.1	2.62	0.4
Montly or annual public transit pass	5.1*	2.35	0.030	21***	3.62	<0.001
Public transit availability in the neighbourhood	6.8*	3.39	0.045	-5.2	3.22	0.11
Indigenous or racialized person	12***	2.20	<0.001	14***	2.71	<0.001
Temporary visa or refugee	-9.1*	3.86	0.019	8.8	5.52	0.11
Person with disability	6.7**	2.39	0.005	-3.2	2.97	0.3
Woman, transgender or non-binary person	2.3	1.96	0.2	-0.47	2.45	0.8
Age	-0.16*	0.081	0.041	-0.21*	0.089	0.017
Household size	-1.3	0.653	0.054	1.4	0.884	0.12
Household income	***		<0.001	***		<0.001
Less than \$40,000	—	—		—	—	
\$40,000 to \$80,000	1.7	2.34		-1.4	2.95	
\$80,000 to \$120,000	-7.5	2.85		-9.2	3.57	
More than \$120,000	-9.9	2.79		-22	3.56	
Population density (people/km2)	0.00**	0.000	0.006	0.00	0.000	0.4
Population centre class	**		0.008	***		<0.001
Rural	—	—		—	—	
Small	-8.5	7.41		11	7.53	
Medium	-18	5.95		-19	5.41	
Large	-1.3	3.86		1.2	4.78	
Active mode interaction: accessibility × user	-12*	5.47	0.027	-4.1	5.48	0.5
Transit mode interaction: accessibility × user	22	15.5	0.2	23	24.7	0.3

Abbreviations: CI = Confidence Interval, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001

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